

Advanced Network & SDN Platform Expert - STAR Case Bank

Comprehensive Profile, High-Priority Incident Management, and Career Guidance (English Version)

I. Core Competencies & Key Highlights Summary

Based on 3.5 years of live-network operations experience servicing major enterprise accounts across China, you possess the following **core competitive advantages**:

- **Full-Stack Vertical Delivery**: Transcends traditional networking boundaries with the capability to manage everything from low-level server hardware (RAID cards, disks, IO, Bonding), Linux OS, containerized microservices, and database table manipulations, up to software-defined network platforms.
- **High-Concurrency Problem Solving**: Consistently resolved **10–15 complex technical tickets per day** under tight deadlines, successfully closing over **12,000+** live-network cases with outstanding multi-tasking and stress-management skills.
- **Data-Driven Operations (Network Programmability)**: Proficient in utilizing Postman, `curl`, and Python scripts to invoke **Northbound RESTful APIs** of iMaster NCE-Campus and eSight platforms for massive data extraction, structural cleaning, and customized data governance.
- **Executive-Level Communication & Collaboration**: Proven ability to take command during major network outages by cross-functionally mobilizing and coordinating a joint task force of **dozens of senior experts** (including R&D engineers, system architects, and third-party vendors) to drive alignment and rapid recovery.

II. STAR Case 1: Full-Stack Network & Software-Defined Network (SDN) Platform Expert

SITUATION

Managed live-network operations and technical support for large-scale enterprise accounts across China (e.g., tier-1 airports, major hospitals, and national retail chains) using advanced network management platforms (e.g., eSight, iMaster NCE-Campus, CampusInsight). Operated as the critical technical point of contact for complex, nationwide infrastructure anomalies and data-driven requirements for over **3.5 years**.

TASK

- **High-Volume Delivery**: Maintain high ticket-closure efficiency to guarantee maximum network availability and business continuity for massive enterprise campus networks.
- **Full-Stack Implementation**: Direct system-level deployment of network management platforms, covering server hardware initialization (RAID/Bonding), network plane architecture, top-level SDN admission control strategies (802.1X/Portal/MAC/HWTACACS), and full-stack microservices/database optimization.
- **Data Governance & Automation**: Overcome the data export constraints of platform graphical user interfaces (GUIs) to pull core underlying live-network data accurately and efficiently for clients.

- **Cross-Domain Orchestration:** Act as the central catalyst during massive, unprecedented cross-system failures that transcend traditional product boundaries to drive rapid problem closure.

ACTION

- **High-Concurrency Execution & Northbound API Governance:** Demonstrated strong multitasking capabilities by ****consistently resolving 10–15 complex technical cases daily over 3.5 years****, accumulating 12,000+ closed cases. To satisfy client requests for audits or network health assessments, ****leveraged the Northbound RESTful APIs of iMaster NCE-Campus and eSight via `curl`, Postman, and Python scripts****. Successfully bypassed GUI limitations to extract massive raw JSON/XML data streams (such as global AP-associated user counts, historical topology link traffic, and AAA authentication failure logs) and performed structural data cleaning to deliver visual executive reports.
- **Bare-Metal Deployment & Full-Stack Optimization:** Guided clients through bare-metal platform installations—ranging from initial RAID card mapping, disk mounting, and OS installation, to the strategic design of Northbound/Southbound management planes and network Bonding for high availability. When facing front-end platform alerts or high CPU utilization in underlying microservices, dived directly into the Linux kernel backend to optimize resource allocation using shell scripts and executed table-level database operations to fix underlying microservice logic anomalies.
- **Advanced Admission Control & Multi-Platform Integration:** Supervised the platform-based management of thousands of network devices via NETCONF and SNMP protocols. Successfully implemented large-scale Fabric/VxLAN architectures and Free Mobility policies on the SDN platform. Solved high-frequency authentication failures involving 802.1X, MAC, Portal, and HWTACACS, and seamlessly integrated network authentication platforms with third-party SSO portals, Feishu, WeChat, and SMS/Email notification gateways.
- **Cross-Domain Technical Leadership (Core Communication):** Broken down institutional silos during highly complex cross-product anomalies. ****As the primary technical commander, successfully assembled and orchestrated online war-rooms comprising dozens of specialists****, including senior R&D engineers and system architects across servers, switches, routers, and WLAN. Unified technical terminology among diverse stakeholders, clearly defined troubleshooting boundaries, assigned diagnostic tasks, and kept the large team strictly focused on core root causes under immense pressure.

RESULT

1. Operational Efficiency: Successfully closed over 10,000 complex cases with zero client complaints or escalations due to communication issues. Spearheaded the resolution of multiple systemic, high-impact industry outages, improving the Mean Time to Resolution (MTTR) for major incidents by over 30%.
2. Data Empowerment: Utilized Northbound API extraction techniques to assist multiple marquee clients in establishing network behavior audits and capacity expansion baselines, substantially maximizing the secondary utility value of network telemetry data.

III. STAR Case 2: Resolving Large-Scale Campus Wi-Fi Outages & Root Cause Isolation

SITUATION

Frequently handled weekly critical, high-priority emergency tickets where major enterprise clients (such as airport check-in systems, hospital registration systems, or massive shopping malls) reported that **entire buildings or designated geographical zones suffered total Wi-Fi service blackouts (users could successfully connect to Wi-Fi but had no internet access)**. These incidents disrupted operations for tens of thousands of end-users, carried immense operational risks, and placed immense SLA and public relations pressure on clients.

TASK

Serve as the primary incident commander for China-wide operations to instantly contain the outage's blast radius, restore critical business flows, rapidly isolate the root cause within a complex multi-vendor SDN ecosystem, and provide objective data-driven evidence for post-incident reviews.

ACTION

- **Rapid Emergency Containment (Business First):** Immediately engaged with client technical teams to assess the blast radius and audit recent configuration changes while managing client anxiety. Prioritizing **business continuity**, if the outage occurred during peak operational hours, I decisively instructed the client to **temporarily modify the wireless/wired authentication policy to "Open" status (bypassing authentication)**, immediately bypassing the failure zone and restoring user internet access within 5 minutes.
- **Baseline Auditing & User Flow Tracking (Trace):** Once business operations stabilized, I applied extensive live-network troubleshooting experience to review configuration baselines across core Portal, AAA, and RADIUS server templates. Simultaneously, I initiated **user trace mechanisms** on the SDN platform and access devices to capture the live end-to-end authentication trajectory of target test terminals.
- **Deep Backend Packet Inspection & Root Cause Isolation:** Penetrated the underlying Linux environment of the SDN platforms and network elements to **execute full packet captures using `tcpdump`**. I meticulously dissected the complete RADIUS handshake sequence (Access-Request, Access-Challenge, Access-Accept/Reject) to analyze exactly where authentication packets were dropped or where credential validation failed. This consistently proved that the vast majority of such outages were caused by intermediary third-party firewalls blocking authentication packets or unauthorized asymmetric configuration changes on the client side.
- **Data-Driven Post-Mortem via Northbound APIs:** During the post-incident review and analysis phase, to provide the client's executive management with an objective impact report, **I utilized Postman and curl tools to directly call the Northbound APIs of NCE-Campus and eSight**. I batch-extracted "global terminal admission failure logs" and "abnormal AP status telemetry" corresponding to the exact timestamp of the outage, performing cross-table data analysis to pinpoint the precise scale of affected users and terminal distributions.
- **Multi-Domain War-Room Mobilization (Core Communication):** For highly elusive bugs embedded in underlying code, intermittent microservice failures, or complex cross-vendor hardware incompatibilities, **I immediately triggered the major incident response framework. Leveraging outstanding technical**

articulation, I mobilized and led a joint task force of dozens of specialists** (WLAN R&D, network architects, server specialists, and third-party vendor teams). I managed the timeline, assigned specific diagnostic tasks, prevented inter-departmental finger-pointing, and ensured a seamless, transparent flow of information between the client and R&D.

RESULT

1. Exemplary MTTR: By leveraging a signature "Open-First Mitigation + Backend Trace/tcpdump Isolation + Northbound API Telemetry" framework, the average recovery time (MTTR) for high-impact, large-scale Wi-Fi outages was kept under 15 minutes.
2. Framework Establishment: The cross-functional war-rooms not only resolved multiple deeply hidden, cross-platform major bugs but also helped clients establish a mature, cross-domain joint troubleshooting SOP, earning high praise from both clients and senior corporate architects.

IV. Recommended Career Paths & High-Compensation Job Roles

Your combined technical acumen qualifies you as a highly sought-after "**Network + Platform + Automation + Linux Backend**" hybrid specialist. It is recommended to target the following roles:

1. **Network Automation Engineer / Network Programmability Expert:** Your proficiency in Northbound API integration, Postman data extraction, and Python scripting provides a generational advantage over traditional CLI-bound network engineers.
2. **Senior Professional Services Consultant / Technical Account Manager (TAM):** Given your extensive background handling 12,000+ tickets and coordinating dozens of R&D engineers, you are exceptionally suited to steer major enterprise accounts and infrastructure projects for global tech vendors.
3. **Site Reliability Engineer (SRE - Network & Infrastructure Focus):** Your deep grasp of bare-metal server deployment (RAID/Bonding), Linux kernel optimization, database management, and large-scale enterprise campus networking provides the ideal structural foundation to transition into high-paying infrastructure SRE positions within top-tier cloud and internet companies.